

QCID-V: The Time Arrow

Irreversibility as a Logical Necessity from the Nothing Paradox

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Version 1.1
September 2026

Abstract

In QCID-IV we demonstrated that 3-dimensional space and 1-dimensional time must co-emerge from Absolute Nothing to resolve the logical paradox of "something from nothing". This paper extends that result. We prove that this emergence cannot be reversible. If emergence were reversible, the net change would be zero and indistinguishable from Nothing. Therefore, Time must possess an intrinsic direction from its origin. We argue that the Second Law of Thermodynamics, the increase of entropy, is not an additional postulate but a direct consequence of the origin of time. From this we derive three falsifiable predictions: the non-existence of white holes, the impossibility of closed timelike curves, and the impossibility of a cosmic "bounce" to a pre-emergence state.

1 Introduction and Recap of QCID-IV

The QCID series seeks first-principles explanations for cosmological structure. QCID-II gave testable predictions. QCID-IV solved the foundation problem.

The core result of QCID-IV: Absolute Nothing cannot contain an event. Therefore, if anything exists, then space and time must have emerged together with it. The minimal stable structure is 3 dimensions of space and 1 dimension of time.

Note: QCID-IV also showed that 3 spatial dimensions are required for stability of orbits and forces. A universe with fewer or more spatial dimensions could not support persistent structures. Therefore the irreversible time we discuss here applies specifically to the stable 3+1 spacetime proven in QCID-IV.

This raises the next logical question: Why does time flow in only one direction? Why do we remember the past and not the future? Standard physics assumes the "arrow of time" as a postulate via the Second Law. Here we derive it.

2 The Reversibility Paradox

Consider two possibilities for the emergence described in QCID-IV.

Case A: Reversible Emergence. Nothing transforms into Spacetime, and Spacetime could transform back into Nothing. Symbolically: $\text{Nothing} \leftrightarrow \text{Spacetime}$.

If this were true, then for every moment of emergence there would exist an equal moment of "un-emergence". Over all time, the net change would be zero. But zero net change is the definition of Nothing. Therefore, reversible emergence is logically equivalent to Nothing.

This contradicts the premise that something exists.

Case B: Irreversible Emergence. Nothing transforms into Spacetime, and there is no path back. Symbolically: $\text{Nothing} \Rightarrow \text{Spacetime}$.

This is the only logically consistent case. Therefore, the act of emergence itself creates a direction. Time is born with an arrow. The past is the direction toward Nothing. The future is the direction away from Nothing.

3 Entropy as a Consequence, Not a Postulate

In standard cosmology, the Second Law is assumed: "Entropy tends to increase." Here we derive it.

Because there is no path from Spacetime back to Nothing, information and energy cannot globally re-concentrate to the origin state. They can only disperse. Dispersion is entropy.

Thus, the increase of entropy is not a separate law of physics. It is the observable signature of the irreversibility of emergence. The universe began in a state of minimal entropy because that state was closest to Nothing. All subsequent evolution must move away from that state.

This explains why the early universe had low entropy without needing special initial conditions.

4 Predictions and Falsifiability

From the axiom of irreversible emergence, we make three concrete predictions:

Prediction 1: No White Holes.

A white hole is the time-reversed version of a black hole. It would require matter to spontaneously re-concentrate from dispersion to singularity. This is a local D to N process and is forbidden by the emergence axiom.

Prediction 2: No Closed Timelike Curves.

A closed timelike curve would allow an observer to return to their own past. This creates a loop where local regions could approach the Nothing state. This violates the global irreversibility.

Prediction 3: No Cosmic Bounce.

Models where the universe collapses and "bounces" back to a new Big Bang require the universe to return toward a pre-emergence state. This is forbidden. The universe can expand, or reach heat death, but cannot reverse to the origin.

5 Connection to QCID-VI and QCID-VII

QCID-VI will show how matter and fields must form within this irreversible 3+1 framework.

QCID-VII will show how an observer, composed of that matter, necessarily experiences the arrow.

6 Conclusion

The arrow of time is not emergent and not accidental. It is foundational. It is the logical price paid for escaping Absolute Nothing. Irreversibility is built into the origin of time itself. Therefore, entropy must increase.

References

- [1] Khan, N. QCID-IV: The Nothing Paradox and the Necessity of 3+1 Dimensional Spacetime. Zenodo. 2026. ORCID: 0009-0002-6006-0796
- [2] Khan, N. QCID-II: Testable Predictions from Quantum Causality. Zenodo. 2026. ORCID: 0009-0002-6006-0796